Part 1: The Distribution of Averages of 40 Exponentials

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Overview

I simulate the exponential distribution (rate $\lambda = 0.2$) and study the mean of 40 draws, repeated 1000 times. By the Central Limit Theorem (CLT) these averages should be approximately normal, centred at $1/\lambda = 5$ with standard deviation $(1/\lambda)/\sqrt{40} \approx 0.79$. I compare the simulated mean, variance and shape with theory. Code is in the appendix.

Simulations

`rexp(40, 0.2)` gives 40 exponentials; I take their mean and repeat this 1000 times (seed fixed for reproducibility). For contrast I also draw 1000 individual exponentials.

```
set.seed(123)
lambda <- 0.2; n <- 40; nsim <- 1000
means  <- replicate(nsim, mean(rexp(n, lambda)))
single <- rexp(nsim, lambda)
```

Sample mean versus theoretical mean

The theoretical mean is $1/\lambda =$ 5; the mean of the 1000 averages is **5.012**, a difference of 0.012. The sample mean of averages is an unbiased estimate of the population mean, so the two agree closely (Figure 1, red = sample mean, blue = theory, almost overlapping).

Sample variance versus theoretical variance

The variance of a mean of n draws is $\sigma^2/n = (1/\lambda)^2/40 =$ **0.625** (sd 0.791). The simulated variance is **0.6** (sd 0.775). They agree to within simulation noise. By contrast, a single exponential has variance 25, so averaging 40 values cuts the variance by a factor of 40.

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Distribution

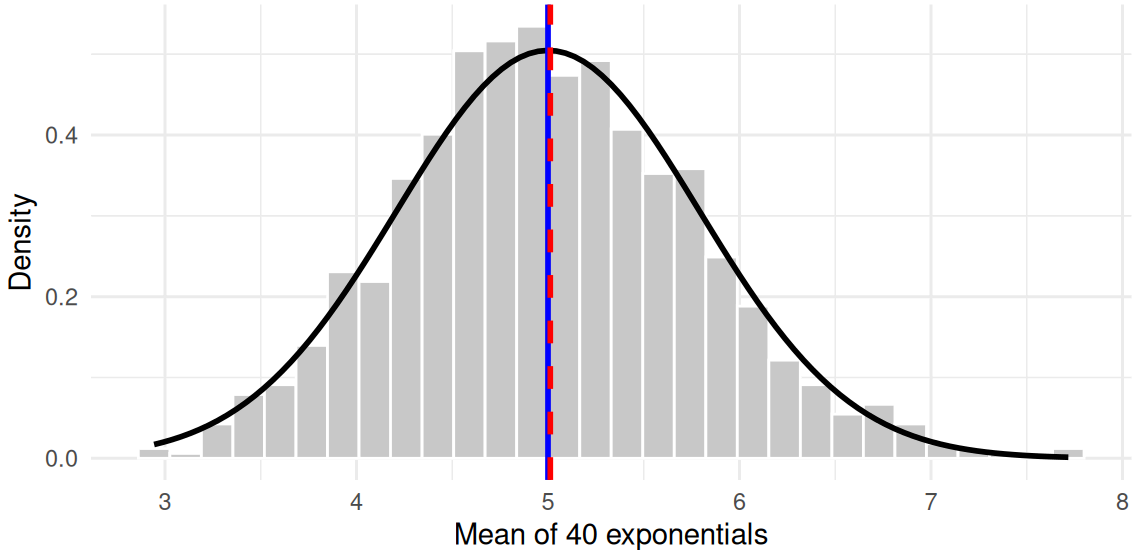


Figure 1: 1000 averages of 40 exponentials, with the sample mean (red), theoretical mean (blue) and the normal density predicted by the CLT (dark curve).

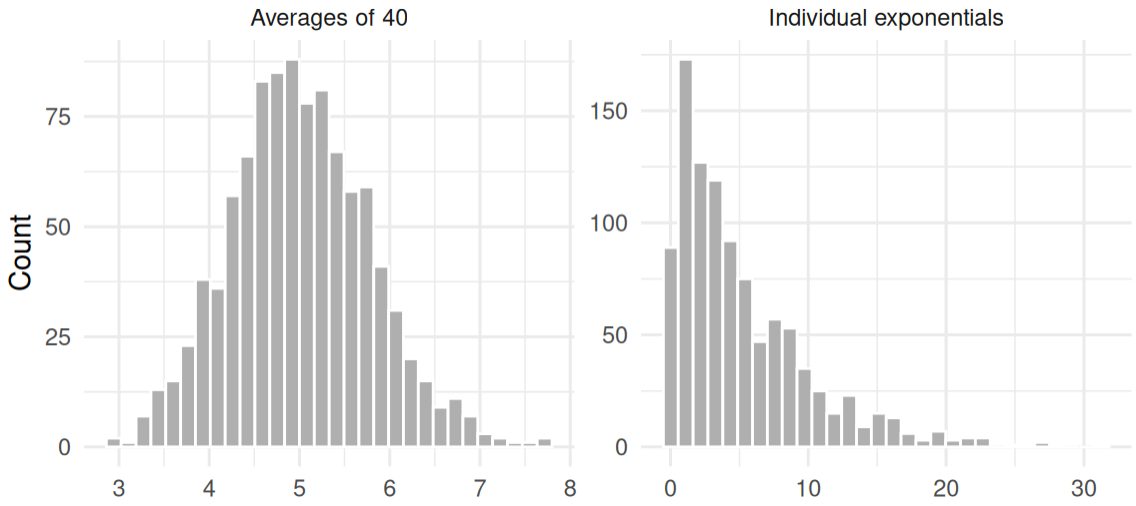


Figure 2: 1000 individual exponentials (left) are strongly right-skewed; 1000 averages of 40 (right) are bell-shaped. See Figure 3 for the Q-Q plot.

The individual exponentials follow the skewed exponential shape (a spike near 0 and a long tail), whereas the averages are symmetric and match the overlaid normal curve. The Q-Q plot confirms this: points lie close to the line except for slight deviation in the extreme tails, where the residual right-skew of the exponential is still faintly visible at $n = 40$.

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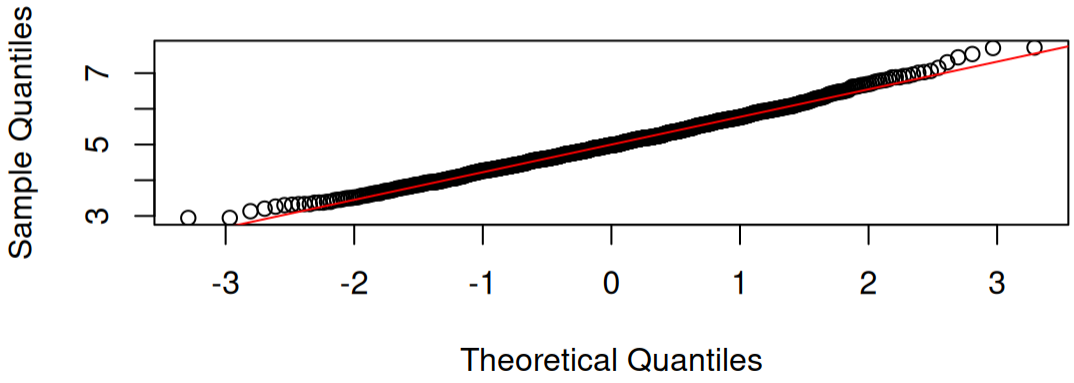


Figure 3: Normal Q-Q plot of the 1000 averages.

As a coverage check, 95.2% of averages fall within ± 1.96 theoretical sd of 5 (normal theory: 95%).

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Appendix: code

```
set.seed(123); lambda <- 0.2; n <- 40; nsim <- 1000
means  <- replicate(nsim, mean(rexp(n, lambda)))
single <- rexp(nsim, lambda)
mean(means); 1/lambda                 # sample vs theoretical mean
var(means);  (1/lambda)^2 / n         # sample vs theoretical variance
hist(means, freq = FALSE); curve(dnorm(x, 5, sqrt(25/40)), add = TRUE)
qqnorm(means); qqline(means)
```

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